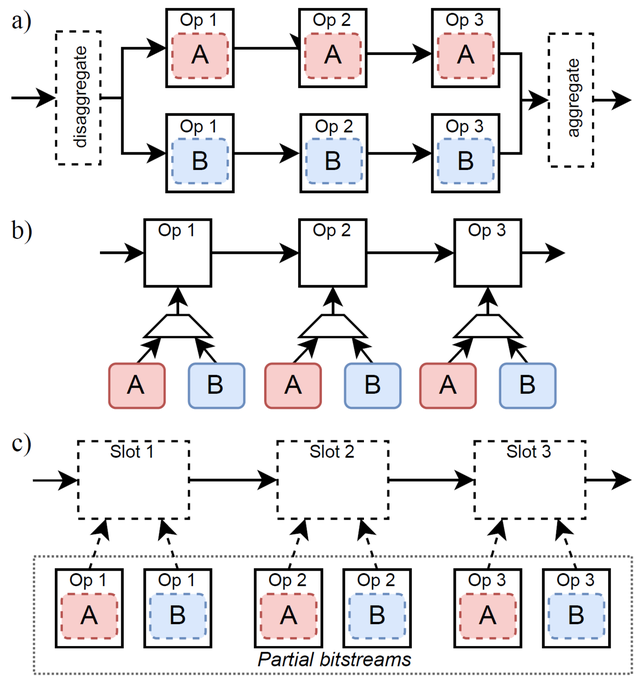
Compiling Multi-DNN Dataflow Accelerators: Dynamic Weight Reconfiguration in FINN

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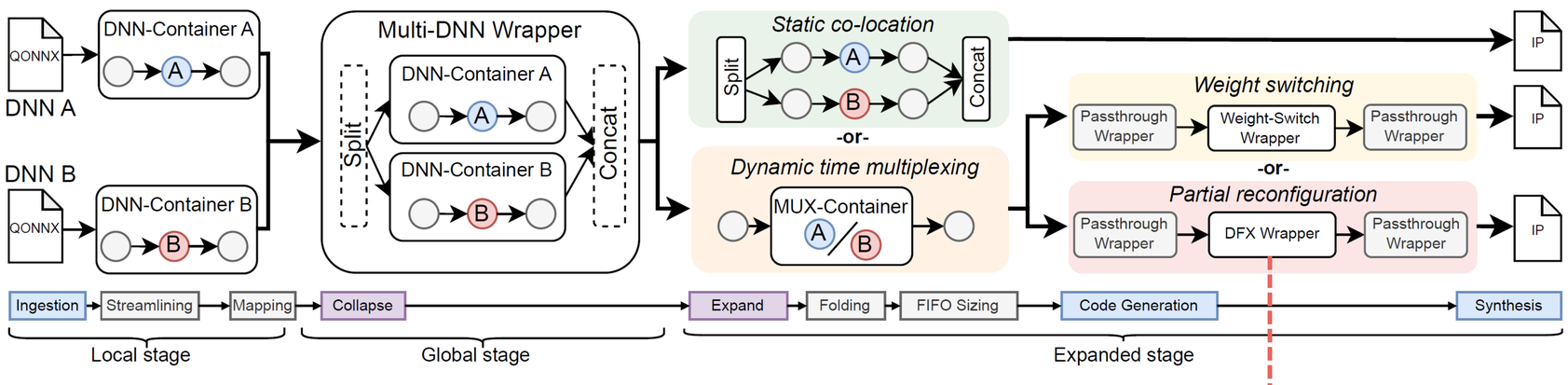
Background

- **Motivation**
  - Multi-DNN workloads run multiple DNNs in parallel to handle distinct tasks or improve result quality
  - Example: Ensemble DNNs, which aggregate the results of multiple DNNs (e.g., via voting)
  - Other advanced DNN topologies could be modeled as Multi-DNN, such as mixture-of-experts or early-exit DNNs
  - Current FINN generates a static accelerator for a single DNN, leaving the FPGA's runtime flexibility unutilized
  - Sharing compute logic can increase resource efficiency due to non-linear resource scaling in FINN
- **Our approach: Share a joint accelerator pipeline among multiple DNNs by dynamically switching or re-configuring between weight sets ("time-multiplexing")**
  - Assumption: Workload consists of multiple DNNs which (at least partially) share a common topology, differing only in weight values
  - Front-end: We extend FINN to ingest a set of DNNs, merging them in a "Multi-DNN Wrapper"
  - Back-end: We support three implementation styles
    - **a) static spatial co-location (baseline)**
    - **b) dynamic weight switching**
    - **c) dynamic partial reconfiguration (DFX)**



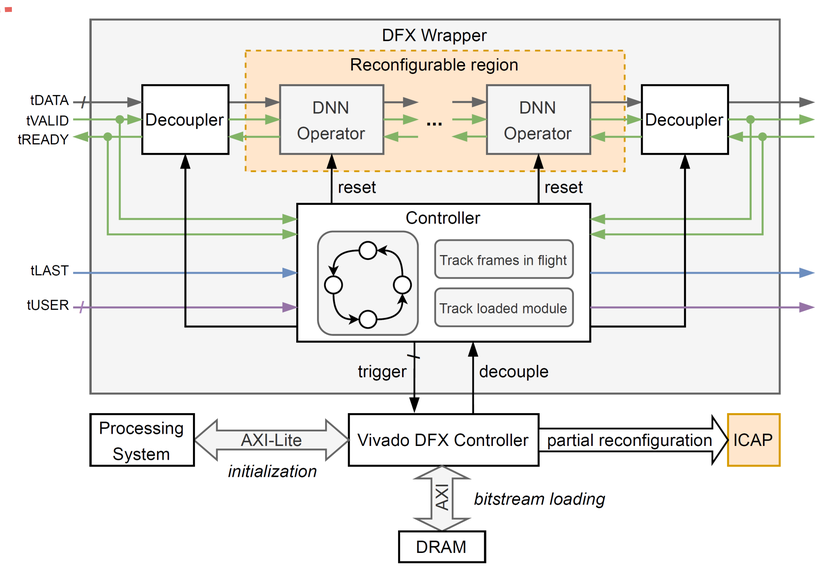
Multi-DNN implementation styles supported by our flow. Example shows two DNNs with three operators. Red and blue boxes represent weight sets of DNN A and B, respectively. Embedding constant weight values into the compute operators is only possible for a) and c).

Methodology



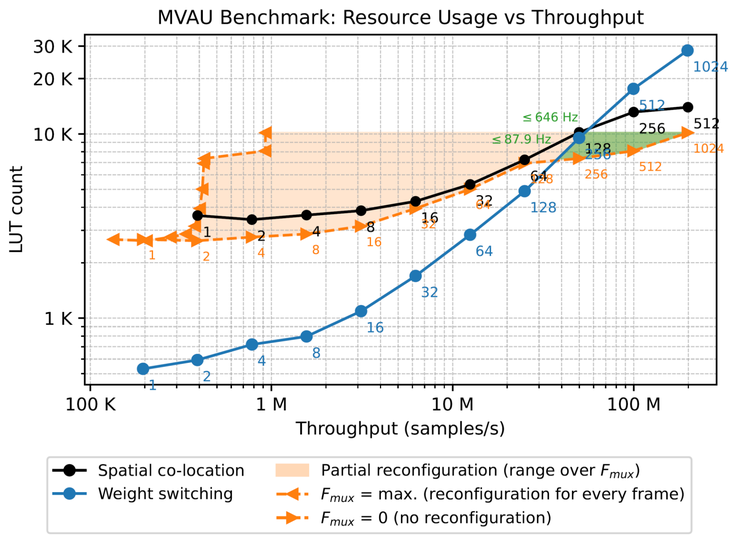
Simplified overview of FINN build flow on an example with two 3-layer DNNs, where a single layer operatores on different weight sets (A and B) and is thus selected for time multiplexing.
Key build steps are annotated below, with modified steps highlighted in blue and new steps in purple.

- **Tool flow** introducing two new meta-level operators:
  - *DNN-Container*: wraps an ingested DNN, steamlining and HW-mapping are performed locally
  - *MUX-Container*: wraps one or more adjacent DNN layers/operators to time-multiplex
  - Top-level graph (Multi-DNN Wrapper) allows us to model auxiliary logic, e.g. split/concat or voting
- **DFX Wrapper component**
  - Wraps operators from a single MUX-Container and places them in a reconfigurable Pblock
  - Interacts with partial reconfiguration infrastructure: Vivado DFX Controller + ICAP
  - Flushes pipeline and decouples AXI-Stream during reconfiguration (opaque from an outside view!)
- **Control scheme** via AXI-Steam side-channel signals:
  - *tLAST* for frame boundary synchronization (handled by wrappers as FINN operators don't use it)
  - *tUSER* for encoding the selected weight set or DFX module (externally driven alongside the primary input stream in our experiments)

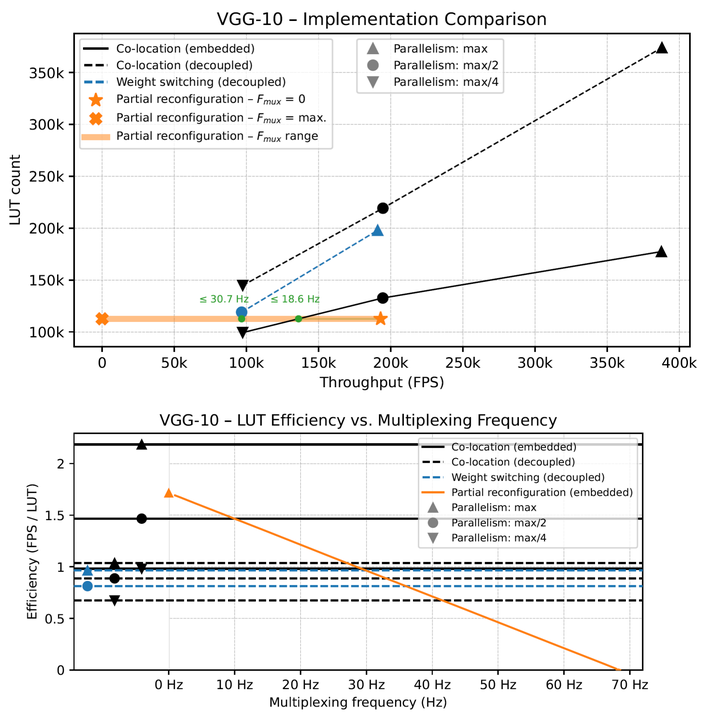


DFX Wrapper implementation. Introduces support for stream sidechannel signals tLAST and tUSER (encodes reconfigurable module selection).

Proof of Concept - Results



MVAU (32×32, 4-bit) benchmark results. Parallelism is swept from minimum (SIMD × PE = 1) to maximum (SIMD × PE = 1024). The throughput in partial reconfiguration mode depends on the time multiplexing frequency Fmux. The Pareto-dominant region over the alternative weight switching implementation is shaded in green, with an annotation of the corresponding min./max. break-even values of Fmux.



VGG-10 DNN results, comparing partial reconfiguration (at max. parallelism) against weight switching and the co-location baseline for multiple parallelism settings.

Future Work

- **Improve automatic floorplanning**
  - Greedy Pblock placement algorithm is coarse-grained and needs Vivado in the loop
  - Resource overhead/fragmentation not yet considered
- **Explore dynamic control ("self-adaptation")**
  - Let the accelerator itself drive the tUSER signal or MLO loop
  - For mixture-of-experts: select expert to load
  - For early-exit: select whether to take exit
- **Explore Hybrid architectures**
  - Combine multiple approaches, for example:
  - *Co-location* for input-specific feature extraction operators
  - *MLO* for repetitive backbone operators
  - *Weight switching* for fine-grained adaptability (per-frame)
  - *DFX* for coarse-grained adaptability (from ~100 Hz to 0 Hz)
- **Find interesting real-world applications**

Link to FINN+ PR

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